AgenticRotation: Learning Local Control for Asynchronous Visual Agents in Semantic In-Hand Reorientation

Anonymous CVPR submission

Abstract

A visual agent can identify a requested object face, but a dexterous hand must maintain contact while the decision is computed. We study semantic in-hand reorientation with asynchronous reasoning and continuous execution. AgenticRotation connects a visual agent to a pose-conditioned student through a common goal representation. Small local controllers adapt joint targets from measured state and request history, using privileged teacher supervision during training. System comparisons cover a marked cube and a die, while control and feedback interventions use only the cube. On the cube, the full Astra system completes 35/40 trials, compared with 30/40 for a bare fine-tuned VLM and 17/40 for a bare rule VLM. Independent confirmation leaves the learned waiting-control advantage unresolved. Continued feedback raises completion from 6/39 for one initial plan to 33/39 with up to six calls. A separate image intervention with the same call cap gives 32/40 with current images versus 9/40 with repeated initial images. In a separate dice layout study, Astra completes 28/40 identity trials and 26/40 after visual-layout rotation. Fine-tuning falls from 23/40 to 0/40 and rules from 13/40 to 1/40. These system-level results support continued visual feedback while exposing contact-control limits. ***[TODO: Complete the multi-object system comparison and physical-hand evaluation.]***

1. Introduction

A command such as "turn the face marked T upward" specifies a visual relation, not a sequence of joint targets. A robot must locate the mark, associate it with a controllable orientation, and move the object without losing it. The requested face may initially be hidden by the object itself or by the fingers. Moving to obtain evidence changes both visibility and contact. This coupling makes in-hand reorientation a useful setting for studying visual agents that act through physical feedback.

Large visual models provide a flexible interface between an instruction and an action. Their inference time introduces another problem. The hand cannot assume that the object remains in place while an external model reasons. Continuing the current skill can expose useful evidence or destabilize the grasp. Slowing every action can reduce movement without supplying the corrective motion needed to recover. A suitable local controller must act on the current contact state even when the next semantic decision is unavailable.

We investigate the following question: *how can a visual agent use successive observations to ground a semantic goal and execute it through continuous dexterous control?* Our system separates the semantic decision rate from the motor control rate. An asynchronous agent requests inspection, binds a visible target, and updates its command from feedback. A history-conditioned student maps a goal representation and measured object pose to joint targets. Two compact learned controllers adapt those targets during pending requests and ordinary execution. This design permits a local policy to react without a second expensive reasoning call.

The visual and control interfaces are explicit. We adapt a ViserDex-style RGB-to-pose policy [1] to a continuous semantic goal input while reusing compatible recurrent weights. A fine-tuned local VLM directly predicts a goal vector. The Astra implementation instead emits validated public tool arguments that are mapped to the same goal space. Thus the controller consumes a common representation, but the Astra branch does not expose hidden model embeddings. The distinction makes it possible to compare a strong external agent with a compact learned grounding model without changing the deployed motor interface.

We freeze the task mapping, checkpoints, stopping path, and physical success rule before evaluation. Each system retains 40 raw trials, with service exclusions applied symmetrically within a comparison. Physical reach is reported separately from correct semantic binding and autonomous completion.

We organize evaluation by scope. Q1 compares complete systems across objects and semantic layouts. Q2 tests control components only on the marked cube, and Q3 uses the same cube to test feedback. Q4 is reserved for multi-object physical-hand evaluation, with a separate cube pose-estimation study as supporting evidence. Current system results cover the cube and die. Historical cohorts remain separate.

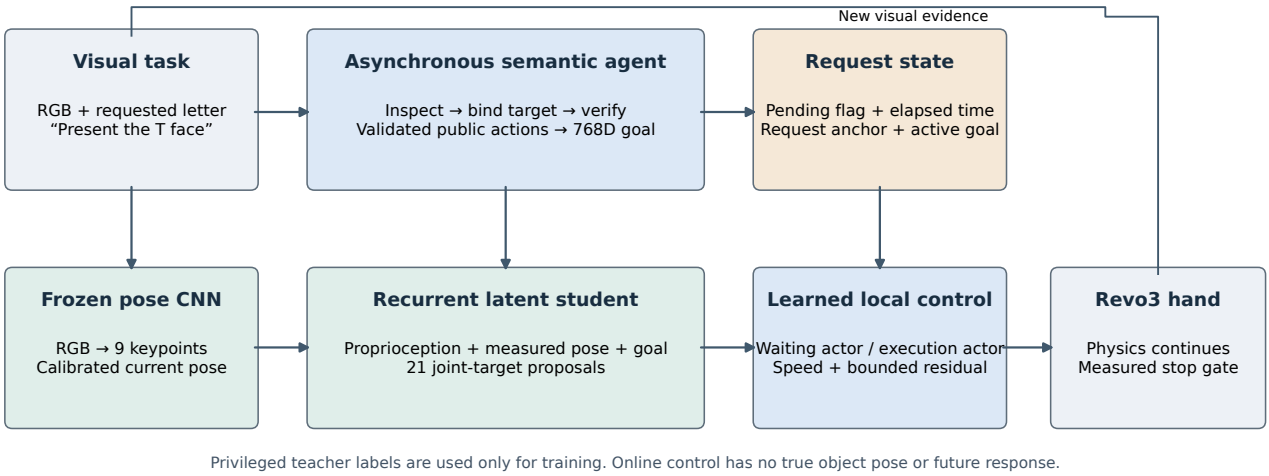


Figure 1. **Continuous control around asynchronous semantic decisions, shown for the cube backend.** The visual agent selects an inspection or target skill. Its validated public command determines a semantic goal vector. A frozen RGB pose estimator and recurrent student run underneath this interface. Separate learned actors adapt joint targets while a request is pending and during execution. The hand continues moving during inference. A common measured-pose gate decides when to stop, and an independent evaluator checks true task completion. The fine-tuned VLM baseline predicts its vector directly and uses the same student without the added actors.

Our contributions are a shared semantic-to-motor interface, teacher-supervised local adaptation during asynchronous reasoning, and matched interventions separating observation, binding and contact failures. The evidence supports continued visual feedback and system-level layout transfer. Independent waiting confirmation, explicit controller-goal benefits and native-layout superiority over fine-tuning remain unresolved. Further objects and hardware validation are necessary.

2. Related Work

Visual dexterous manipulation. ViserDex uses monocular visual pose estimation and recurrent control for dexterous reorientation, with Gaussian representations supporting visual training [1]. We reuse this division between perception and control and adapt a released Revo3 policy to semantic goals. ManipTrans studies residual learning for dexterous bimanual transfer [4]. These works motivate a reusable motor policy and residual adaptation. Our residual parameterization alone is not a novelty claim. The question here is how a local policy should condition its behavior on the phase and intention of an asynchronous semantic agent.

Language and visual agents. Magma connects multimodal representations to agent actions [10]. VERA translates visual plans into robot actions and includes language-conditioned dexterous in-hand reorientation [5]. Its hand experiments include directional rotation prompts. Consequently, language-guided in-hand rotation is not an unoccupied task category. Our setting emphasizes identifying a requested marked face under occlusion, maintaining contact during external reasoning, and checking a canonical target at autonomous stopping. Those distinctions require direct evidence, rather than a claim of being the first language-controlled hand.

Asynchronous robot policies. Real-time chunk execution [2], online corrections for VLA chunks [8], and latency-aware RL [11] address inference delays through different execution and learning designs. DEFLECT studies delay-robust asynchronous VLAs through temporal counterfactual preferences [12]. LiMA combines sparse semantic intention and asynchronous action generation for dexterous manipulation [3]. These are close methodological references. Our current implementation learns small teacher-supervised controllers around a separate recurrent policy and public semantic tools. A controlled comparison with a generic time-aware residual policy is still needed to show that the active semantic goal and request anchor add value beyond reactive correction.

3. Task and Evaluation Contract

Semantic target. The marked cube is the common object for all Q2/Q3 interventions. Q1 also includes a wooden die with instructions requesting one through six dots, with further objects pending. Cube instructions identify one letter from $\{T, E, R, L, S, H\}$. Each face has a fixed color and a canonical world orientation $Q(g)$ that places its normal upward. The environment scores the full canonical quaternion, including its fixed twist, rather than accepting every

orientation with the correct normal. The language prompt is therefore less specific than the scoring convention. This convention is disclosed and held constant for all methods. A face-normal-only score can be reported as an additional diagnostic without replacing the primary metric.

At step t, the agent observes RGB evidence I_t and the instruction ℓ. The motor system observes proprioception x_t and measured pose $\hat{p}_t$. True pose p_t, teacher actions, and future semantic responses are unavailable to online control. The simulator exposes them only for training labels and independent scoring. We denote the active goal vector by g_t, recurrent state by h_t, and the pending-request flag by m_t.

Physical time. One request may be pending at a time. The default protocol allows six model calls per episode and 3,600 logical control steps at 30 Hz. If a call starts at k and takes L wall-clock seconds, its response becomes usable only after the call completes and at least $\lceil 30L \rceil$ control steps have elapsed. Physics continues throughout this interval. The simulator currently runs slower than real time. This protocol represents inference delay in simulated physical time, but does not demonstrate a 30 Hz hardware pipeline.

Autonomous completion. All cube systems share a measured-pose stopping gate, evaluated at 3 Hz. It requires a declared binding for the requested symbol, no pending request, a non-inspection state, and measured full-orientation error at most $\epsilon = 0.16$ radians. The gate may stop independently of an agent's native stop suggestion. The dice extension uses a calibrated completion correction for Astra and a measured-pose gate for the bare baselines, with the same independently scored tolerance; its protocol is documented separately in the supplement. At stop time τ, the independent success variable is

$$S = \mathbf{1}\left[\hat{b}_\tau = b^*(\ell),\ d(q_\tau, Q(b^*)) \le \epsilon,\ \neg D_{0:\tau}\right], \quad (1)$$

where $\hat{b}_\tau$ is the declared face binding, q_τ is true orientation, $D_{0:\tau}$ indicates an earlier terminal drop, and $d(q, Q) = 2\arccos(|\langle q, Q\rangle|)$ for normalized quaternions. Merely passing through a target does not count. No extra hold duration is added after stopping. A pose false stop and a semantically wrong binding are failures even if the hand retains the object. The frozen drop rule is distance of at least 0.24 metres from the nominal object grasp position.

Matched reporting. Every arm uses 40 raw trials from two seeds, with 20 trials per seed and matching initial physical states and scored physical targets. Instructions also match within the control and feedback studies. In the remapping study, the requested letter changes with the mapping to preserve the physical target. A service error or measured model latency above 120 seconds is recorded as a service exclusion. For a panel of methods $\mathcal{C}$, the eligible set removes the union of excluded trial identifiers across $\mathcal{C}$. We report both raw and eligible outcomes, never replace an excluded trial, and never mix denominators from different panels. Long service delays remain visible without being interpreted as controller failures.

4. Method

Figure 1 summarizes AgenticRotation. The system has a semantic interface, a visual motor policy, and local adaptation. The semantic interface changes the active intention. The local controllers change how that intention is executed using current measurements.

4.1. Reusing a visual pose-conditioned student

A frozen CNN predicts the image coordinates and depth of the object center and eight keypoints from a 120×120 RGB image. Calibration, unprojection, and rigid fitting recover $\hat{p}_t \in \mathbb{R}^7$, comprising position and an XYZW quaternion. The fit residual accompanies the measurement. The current rendering uses the object's Gaussian representation with live black hand occlusion. It is not a full-scene image or a real-camera observation, and no ground-truth pose fallback is used at deployment.

The inherited student expects geometric goals. We replace its goal channels with a learned projection of $g_t \in \mathbb{R}^{768}$:

$$z_t = \tanh(W_g g_t + b_g) \in \mathbb{R}^{64}, \quad (2)$$

$$(u_t, h_{t+1}) = \pi_\theta([x_t, z_t], \hat{p}_t, h_t). \quad (3)$$

Here $x_t \in \mathbb{R}^{109}$ contains target-free proprioceptive features and history, while $u_t \in \mathbb{R}^{21}$ gives normalized joint-target proposals. Compatible encoders, recurrent weights, actor weights, and normalization statistics are initialized from the released policy. Neither target quaternion nor relative-goal quaternion is an input to the migrated student.

The first training stage fits the new interface while preserving inherited coefficients. A second stage adapts the recurrent core using actions labeled by the frozen privileged teacher. Teacher and student rollouts are mixed in a DAgger-style procedure [7]. The frozen release uses 200 interface updates and 400 adaptation updates, each with 128 environments and 45 rollout steps, totaling 3,456,000 transitions. These stages use simulated pose noise. All reported cube closed-loop trials instead obtain pose from the frozen CNN. We separately measure the loss introduced by this transfer rather than assuming that initialization preserves the teacher's performance.

Dice backend. The second object reuses a released feedforward policy over ten observation frames. A learned

adapter maps continuous goal history and measured orientation to eight normalized conditioning features for that frozen actor. It receives no target quaternion at inference. This backend uses the released noisy and delayed simulator pose model, not a visual predictor. Its maintenance and calibrated completion modules are fitted separately. The supplement records these differences; the dice comparison tests an object-specific system extension, not zero-shot transfer of cube weights.

4.2. Semantic grounding and public actions

We maintain six normalized goal vectors $B = \{g_1, \dots, g_6\}$ from cached SigLIP2 text features [9]. Each vector denotes a color skill and is bound to a canonical orientation in the audited asset. The goal bank is fixed for the cube release.

Astra interface. Astra receives the requested letter and visual observations. It emits a validated public action from `inspect_face`, `reorient_target`, `verify_target`, `stop`, or `abstain`. A color argument selects a vector from B. Inspection can change the motor goal without declaring that the requested letter has been found. Inspection actions can request between 10 and 90 control steps. Subsequent visual evidence and public action history support another decision. We use the name Astra for the hosted route `gpt-6-astra` recorded in our experiment, not for a model trained here. Its exact reproducible provider version is a release requirement.

The vector delivered to the student is a local encoding of a public command. We do not access Astra's internal features, train Astra weights, or describe this pathway as end-to-end language-to-action learning. This explicit interface also enables fixed-public-plan diagnostics in which the motor system is evaluated under recorded decisions and delays.

Fine-tuned VLM. The compact baseline uses Qwen3-VL-4B-Instruct [6] with low-rank adaptation and a goal head. The head predicts a normalized continuous vector in the centered five-dimensional span of B, together with a readability probability. The continuous vector enters Eq. 3 directly. Its nearest canonical vector determines the declared binding and the orientation used by the shared stopping gate. This interface is not unrestricted open-vocabulary motor generation.

Let y indicate whether the requested mark is readable, k its skill index, and $\hat{g}$ the predicted goal. Training combines readability binary cross-entropy with positive-example goal classification and cosine alignment:

$$\mathcal{L}_{\text{ground}} = \text{BCE}(\hat{y}, y) + \mathbf{1}[y = 1]\big(\text{CE}(\hat{k}, k) + 1 - \cos(\hat{g}, g_k)\big). \tag{4}$$

The implementation averages positive terms over readable examples. Readability supervision uses geometric visibility and an image-signal filter, not human judgments. Training first uses teacher-view data and then includes observations from student trajectories. The current objective does not include an action-consistency loss.

Rule VLM. The rule baseline parses a color and readability decision from the local VLM and maps accepted colors to B. Both local baselines use the same fixed inspection sequence when a view is not accepted. They use the migrated student directly, with no learned waiting actor, execution actor, or learned completion correction. This is the primary complete-system comparison. A separate auxiliary comparison adds the same control actors to each provider to examine the control contribution.

4.3. Learning control while a request is pending

An outstanding query does not determine whether the current movement is still safe. We construct a deployable feature vector s_t from proprioception, measured pose, student belief, proposed action, pose-fit residual, semantic binding status, and request status. It also includes the measured pose and command at the request anchor and elapsed pending time. The active goal g_t is encoded separately. Before a binding is established it denotes the startup or inspection skill, not the unavailable future target.

A 192-dimensional sensor encoder and 64-dimensional goal encoder feed a 128-dimensional fusion layer. A waiting actor produces one speed logit r_t and a residual $v_t \in \mathbb{R}^{21}$. Let $\bar{q}_t$ be normalized current joint position and $\bar{u}_t$ the clipped student proposal. The applied command is

$$\alpha_t = \text{sigmoid}(4r_t + \text{logit}(0.15)), \tag{5}$$

$$b_t = \begin{cases} \bar{q}_t + \alpha_t(\bar{u}_t - \bar{q}_t), & m_t = 1, \\ \bar{u}_t, & m_t = 0, \end{cases} \tag{6}$$

$$a_t = \text{clip}\big(b_t + 0.35\tanh(v_t), -1, 1\big). \tag{7}$$

Zero outputs reproduce fixed slowing with a coefficient of 0.15 during a pending query. The coefficient scales the proposed joint-position displacement while simulation time and the control loop continue to advance. The environment applies its original joint mapping and smoothing. Scalar slowing restricts the command to the segment between current joint position and the student proposal. The joint residual can change the command outside that segment, which permits contact adjustments that a speed coefficient alone cannot express.

Waiting targets are generated by the frozen teacher with the request anchor as the holding objective. This supervision is available in simulation. The deployed actor receives measurements and the active goal only. It never receives

the answer being computed. The objective fits the applied action to the teacher label and penalizes residual magnitude:

$$\mathcal{L}_{\text{control}} = \mathbb{E}_{(s,g,a^*)\sim\mathcal{D}} \left[\|a_\phi(s,g) - a^*\|_2^2 + 0.01\|\delta_\phi(s,g)\|_2^2 \right]. \quad (8)$$

Refitting the waiting policy. The waiting actor is refitted for 2,400 updates on separately collected measured-state trajectories. Each actor batch contains one half pending states and one quarter each unbound and bound execution states. A separate fixed-plan development gate selects uniform pending sampling over a startup-balanced candidate. Only pending-request outputs are deployed. Execution, student, predictor, and stopping remain fixed. The supplement records selection and fresh live confirmation. Neither uses the subsequent three-system or Q2/Q3 test outcomes.

4.4. Learning execution after a decision

A second actor with the same architecture serves non-pending execution. It uses the student proposal as its base, with no speed attenuation, and learns the bounded correction in Eq. 7. Its target is the teacher action for the active goal. It can remain active throughout execution, including inspection, rather than only during a short handover interval.

Paired training examples reuse a physical state and recurrent history with different valid goals. Features and teacher labels are reconstructed for each goal. The motivation is that a generic holding action can preserve contact without moving toward the requested orientation. Goal-dependent labels give the actor a reason to retain directional information. This data construction does not itself demonstrate that the deployed network uses semantics. We test this mechanism with a matched no-explicit-goal actor, shuffled-goal evaluation, and restored-state interventions. The completed tests do not establish a benefit from the explicit actor-goal pathway (Sec. 5).

Training also includes states collected under recorded public action and delay schedules, matching the controller's pending and execution phases more closely to deployment. The selected execution actor is the v6 candidate from development. The same weights are retained throughout the frozen evaluation. The original development selection was informed by observed development outcomes, not a blind preregistration. Reserved evaluation results did not select the checkpoint. Learned completion heads were explored during development but are not active in the reported release.

4.5. Closed-loop execution

At each control step, the system updates visual pose, advances the recurrent student, chooses the waiting or execution actor, and applies the resulting joint targets. A completed semantic response is validated before it changes the active skill or declared binding. The next image reflects the motion that occurred while the previous response was pending. The common stopping gate is checked separately from semantic action generation. Invalid model output cannot supply ground-truth information or silently select the requested target.

Student adaptation and controller learning are evaluated separately. A new student changes the belief and proposal distribution seen by the controllers. Their checkpoint-bound weights therefore require compatibility validation before transfer.

5. Experiments

We ask four questions: **Q1** How do complete systems compare across objects and semantic mappings? **Q2** How should the hand act during deliberation? **Q3** Does continued visual feedback help? **Q4** Does the system work on physical objects? Multi-object coverage belongs to Q1 and Q4. All Q2/Q3 interventions use only the marked cube to keep assets, perception and motor control fixed. Q1 currently covers two objects. Further objects and hardware remain TODO.

5.1. Setup and reporting

Experiments use Revo3 cube and dice assets in Isaac Lab 3.0/Newton with validated contact capacities. Each agent condition has two seeds and 40 scheduled trials. Initial physical states are matched within each study. Within the cube studies, all systems share the frozen pose estimator, goal bank, latent student, 0.16-radian success threshold, and measured-pose stopping gate. Dice trials share an object-specific student and noisy simulator pose; Astra additionally uses calibrated completion correction. We report these backend differences rather than pooling their protocols. The primary fine-tuned and rule VLM baselines omit the added waiting and execution actors. Astra uses adaptive inspection, whereas the local baselines use a fixed unreadable-view schedule. This is a comparison of complete systems, with natural model latencies, rather than an isolated test of model weights.

The latest cube system replaces only its waiting actor. Release cohorts and module interventions remain separate. Table percentages use common service-eligible trials within the stated comparison. All raw records are retained, and no failed or excluded slot is replaced. Wilson intervals describe individual rates. The new waiting contrasts and six registered visual contrasts use Holm correction within their respective families. The independent confirmation and image-refresh tests form a separate two-test Holm family. Historical paired tests remain exploratory and unadjusted.

System	Successes	Rate (%)	95% Wilson interval	Drops	Model calls
Astra full	35/40	87.5	[73.9, 94.5]	3	101
Fine-tuned VLM, bare	30/40	75.0	[59.8, 85.8]	9	117
Rule VLM, bare	17/40	42.5	[28.5, 57.8]	12	126

Table 1. **Q1: latest native-image cube comparison.** Each system has 40 matched trials, with no service exclusions. Astra uses refitted waiting control and the original execution actor. The two VLM baselines use the shared student without these actors. Model calls total all 40 trials. This evaluation uses the original object/hand RGB pipeline, not the reconstructed images used for the separate semantic intervention.

Simulation object	Rule	Fine-tuned	Full
Cube, latest release	17/40	30/40	35/40
Die, noisy pose	17/40	20/40	24/40
Object 3	TODO	TODO	TODO
Object 4	TODO	TODO	TODO
Object 5	TODO	TODO	TODO

Table 2. **Q1: object coverage.** Cube and die rows are measured under their own frozen releases. Dice uses noisy simulator pose without a predictor. Further objects require recorded geometry, calibration, policy, goal definitions, and 40 matched trials per method. Object-specific students do not imply zero-shot transfer. No average is computed over unfinished rows.

System	Identity	Permuted	Masked
Astra full	32/40	32/40	0/39
Fine-tuned VLM, bare	30/40	16/40	3/40
Rule VLM, bare	23/40	23/40	0/40

Table 3. **Q1: cube semantic-mapping comparison.** Identity and permutation retain all 40 trials. Astra masking has one service exclusion. Its paired masking contrast uses 31/39 identity successes versus 0/39 masked successes. The local methods have no exclusions. This study uses the original Astra actors.

5.2. Q1: Systems across objects and mappings

The latest full Astra system succeeds in 35/40 trials (87.5%), compared with 30/40 (75.0%) for the bare fine-tuned VLM and 17/40 (42.5%) for the bare rule VLM (Table 1). Astra has nine exclusive successes against four for the fine-tuned VLM, giving exact paired $p = 0.267$. The 12.5-point difference is therefore a favorable estimate, not established superiority. Against the rule VLM, the corresponding counts are twenty and two ($p = 0.000121$).

Remaining failures and prior releases. The latest Astra cohort has three drops during unbound inspection, one wrong-color binding followed by a stop, and one abstention. No pending-stage drop occurs in this cohort, although Q2 contains such failures. Fine-tuning has nine execution drops and one horizon failure. Rules have twelve drops, ten false stops and one horizon failure. The supplement retains earlier cohorts, whose rankings also vary, without pooling controller versions. Timing logs confirm continued control during pending requests and delayed reply application, not real-time simulator throughput.

Second-object extension. The native-layout die yields 24/40 successes for Astra, 20/40 for fine-tuning, and 17/40 for rules, with no service exclusions. Astra versus fine-tuning remains unresolved (exact paired $p = 0.4545$). Twelve of its fifteen drops occur before the first skill arrives. This transfer therefore exposes a startup-control limitation despite successful semantic binding. The weights and perception backend differ from the cube, so this row does not isolate a geometry effect. The separate visual-layout study below uses a shared physical-skill prompt and is not pooled with this native-layout result.

Visual-semantic intervention. We keep predictor input and lighting fixed while changing the upper-level image. Identity, one cyclic letter–color permutation, and letter masking are evaluated on 40 matched states per method. All three conditions use geometry-consistent reconstructed face images, including identity. Consequently, this is a controlled image benchmark rather than the untouched RGB main comparison.

Astra retains 32/40 successes after permutation, while fine-tuned success falls from 30/40 to 16/40 (Holm-adjusted $p = 0.0130$ across six planned contrasts). Rule success remains 23/40. Equal aggregate rates do not establish equivalence or identical trajectories. Masking removes almost all completion, supporting dependence on visible semantic evidence. Nine of eleven fine-tuned wrong-binding trials after permutation select the original letter–color association. This is consistent with reliance on learned associations, but does not identify an exclusive memorization mechanism. One cube and one permutation do not establish broad semantic generalization.

Dice layout transfer. A separate 240-trial study rotates only the native visual mesh, preserving collision and inertia.

Waiting policy	Success	%	Drops	Wait drops
Continue student	25/40	62.5	12	7
Fixed slowdown	30/40	75.0	4	4
Learned slowdown	28/40	70.0	8	1
Learned slowdown + correction	34/40	85.0	5	4

Table 4. **Q2: cube controller ablation.** Each condition has 40 scheduled trials. All arms use the same frozen controller outside pending requests. Successes and drops use the common service-eligible panel. Wait drops combine startup and later pending intervals. All 40 trials per arm are service-eligible.

The same two layouts use a prompt distinguishing printed counts from motor-skill IDs. With 40 matched trials per cell, Astra completes 28/40 identity and 26/40 permuted trials, fine-tuning 23/40 and 0/40, and rules 13/40 and 1/40. There are no service exclusions. In the permuted condition, Astra's exploratory paired contrasts give $p = 2.98 \times 10^{-8}$ against fine-tuning and $p = 5.96 \times 10^{-8}$ against rules. Its own layout contrast ($p = 0.688$) does not establish equivalence. The local baselines receive current RGB and instruction, whereas Astra also uses probes, history and noisy feedback. This result supports system-level layout transfer, not isolated model superiority. The supplement reports protocol differences, binding errors and a recorded probing example.

5.3. Q2: Controller ablations on the cube

On the marked cube, we compare four ways of acting while a semantic request is pending: continue the student proposal, interpolate toward it with fixed speed 0.15, learn that speed, and learn both speed and a bounded joint correction. Every condition uses the same frozen execution actor outside pending intervals. Astra, the student, visual pose predictor, stopping gate, and physical horizon are fixed.

The speed-only actor is trained with zero waiting residual, matching full-actor data, initialization, optimizer, sampling and 2,400 updates. Reproduction yields identical deployed full-actor tensors. Held-out checks verify identical non-pending actions.

Continued execution succeeds in 25/40 trials, fixed slowing in 30/40, learned slowing in 28/40, and learned slowing with joint correction in 34/40 (Table 4). There are no service exclusions. The full actor has the highest observed success, but learning only the speed does not improve on fixed slowing. Holm-adjusted exact paired p-values for the three registered adjacent contrasts are 0.604, 0.804, and 0.539. This cohort does not resolve those component gains.

Total drops are twelve, four, eight, and five. Full control has four later-pending drops versus one for learned slowing. Fixed slowing has four startup drops and no later drops. Highest completion therefore does not coincide with fewest drops. The supplement separates phases. Shared execution weights do not imply identical states or subsequent agent decisions.

Protocol	Success	%	Drops	Calls
One initial plan	6/39	15.4	0	40
Full feedback	33/39	84.6	5	120

Table 5. **Q3: initial plan versus feedback on the cube.** Each arm has 40 scheduled trials. The full-feedback condition is shared with Q2, rather than counted as another 40 raw trials. Calls include all raw episodes. One service-affected slot is excluded symmetrically. Different inference budgets are part of the protocol comparison.

Trials use seeds 185501 and 185502 with reversed condition order for the second seed. Checkpoints are fixed before these outcomes. Historical training-version comparisons remain separate from this structural ablation.

Independent confirmation. Without refitting, a new two-seed cohort gives 25/39 successes for continuation, 31/39 for fixed slowing, and 30/39 for learned slowing with correction. One service slot is excluded symmetrically, with all 120 raw trials retained. Learning versus fixed slowing has six versus seven exclusive successes (Holm $p = 1.0$). The learned arm has no waiting drops but eight execution drops, compared with three and two for fixed slowing. This confirmation does not establish a learned-control advantage. Later failures may reflect states and commands induced by waiting, despite identical execution weights. The supplement reports the full panel.

Other component evidence. Earlier waiting–execution factorial tests do not establish a reliable interaction. Explicit controller-goal interventions also do not establish a benefit, and a separate learned execution-residual gate fails its development admission rule. These negative results and the matched-state diagnostic are retained in the supplement. The new waiting study changes neither the execution actor nor its goal input.

5.4. Q3: Visual feedback on the cube

On the same cube, we compare Astra under two observation protocols. One receives the initial RGB observation and measured feedback, then commits to a complete plan containing one to six bounded actions. It receives no later semantic observation. Its last non-abstain skill can continue until the shared stopping gate or horizon terminates the trial. The other can use up to six calls with updated observations and history. Both retain continuous pose feedback to the student, the same full waiting and execution actors, and the same stopping rule.

The initial plan succeeds in 6/39 eligible trials, compared with 33/39 for continued feedback. Twenty-seven paired tri-

Upper RGB	Success	%	Drops	Calls
Current image	32/40	80.0	3	112
Repeated first frame	9/40	22.5	4	98

Table 6. **Q3: fresh visual evidence on the cube.** Forty trials per arm, with the same call cap and frozen local controllers. Only upper-level RGB is frozen. Calls are realized totals, not matched compute.

als succeed only with feedback and none only with the initial plan (exact $p = 1.49 \times 10^{-8}$). Raw success is 6/40 versus 34/40. One initial-plan HTTP 503 timeout is excluded from both arms without replacement. Actual calls total 40 and 120.

The initial plan abstains in 28 eligible trials, of which 22 succeed with feedback. Among eligible feedback trials, 36 contain inspection, 32 first bind after inspection, and 31 successes use multiple calls. These traces support obtaining evidence before binding, without demonstrating correction of an executed wrong binding. The protocols differ in inference budget. Zero initial-plan drops accompany frequent abstention, not better completion. The supplement retains a recorded example and the earlier cohort.

Does the next image matter? A separate cohort retains the same six-call cap and trigger policy in both arms, changing only the upper agent's image. One arm receives current RGB, the other the exact first frame on every call. Numeric pose feedback, action history, lower visual perception and control remain live. Forty pairs share initial physical states and first images, with no service exclusions.

Current images yield 32/40 completions versus 9/40 for repeated first frames (Table 6). All 23 discordant pairs favor current images, a 57.5-point gain (exact $p = 2.38 \times 10^{-7}$, Holm $p = 4.77 \times 10^{-7}$). Abstentions fall from 24 to two, while drops change from four to three. Transport hashes verify the delivered images. This supports the value of later visual evidence with other feedback retained. Actual calls, tokens and trajectories differ, so equal maximum calls do not establish equal realized compute.

5.5. Q4: Multi-object hardware evaluation

Hardware protocol. **[TODO: Insert multi-object hardware results.]** Compare full Astra, bare fine-tuned VLM and bare rule VLM on each physical object, with 40 trials per method across two sessions. Report per-object completion, drops, false stops and inference cost, using independent scoring and synchronized videos. Record object-specific policies and calibration. Hardware results remain pending.

Supporting cube perception study. A paired study evaluates fixed RGB and RGB-D estimators on 6,144 rendered cube observations across three seeds and four camera pitches. Recall requires translation error < 10 mm and rotation error $< 10^\circ$, with invalid predictions counted as failures. RGB-D reaches 3,977/6,144 (64.73%) versus 2,306/6,144 (37.53%) for RGB. The paired gain is 27.20 percentage points (95% environment-cluster bootstrap interval 26.07–28.30). At 25–50% occlusion, recall is 55.45% versus 19.91%, but both reach zero at $\geq 75\%$. The supplement gives stricter thresholds, no-hand controls and depth perturbations.

Training histories differ, and evaluation uses ideal target masks and virtual depth. This supports a perception-system comparison under light and moderate occlusion, not an isolated fusion effect. These checkpoints differ from the frozen agent predictor. Downstream task gains and physical-camera accuracy remain unmeasured.

6. Discussion and Limitations

Q2/Q3 examine mechanisms only on the marked cube, so their effects are not established across objects. The image-refresh intervention retains numeric feedback, history and lower perception. Its call cap is matched, but realized compute and trajectories differ.

The earlier waiting study favors joint correction by point estimate, but independent confirmation gives 30/39 versus 31/39 for fixed slowing. Zero waiting drops in that learned arm coexist with eight execution drops. A completion advantage remains unconfirmed. Explicit controller goals provide no demonstrated benefit. The rejected execution gate is not deployed. Native-layout Astra–fine-tuned differences remain unresolved on both objects.

Q1 tests one changed mapping per object, with system differences in history, feedback and control. Cube perception uses rendered RGB, while dice uses noisy simulator pose and separate weights. These results establish neither real-camera robustness nor zero-shot object transfer. Hosted-model revisions limit replication. Predictor accuracy does not establish downstream gains. Further Q1 objects and Q4 hardware results remain pending.

7. Conclusion

We studied semantic in-hand reorientation with continuous control around asynchronous visual decisions. Cube interventions support continued feedback, and system comparisons support layout transfer on the tested cube and die. Component gains remain unresolved, and dice startup drops expose a remaining control weakness.

References

[1] Arjun Bhardwaj, Maximum Wilder-Smith, Mayank Mittal, Vaishakh Patil, and Marco Hutter. ViserDex: Visual sim-to-real for robust dexterous in-hand reorientation. *arXiv preprint arXiv:2604.11138*, 2026. 1, 2

[2] Kevin Black, Manuel Y. Galliker, and Sergey Levine. Real-time execution of action chunking flow policies. *arXiv preprint arXiv:2506.07339*, 2025. 2

[3] Ning Chen, Yankai Fu, Junkai Zhao, Qianpu Sun, Guocai Yao, Pengwei Wang, Zhongyuan Wang, and Shanghang Zhang. LiMA: Bridging long-term imagination to real-time dexterous manipulation via asynchronous diffusion. *arXiv preprint arXiv:2609.28431*, 2026. 2

[4] Kailin Li, Puhao Li, Tengyu Liu, Yuyang Li, and Siyuan Huang. ManipTrans: Efficient dexterous bimanual manipulation transfer via residual learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6991–7003, 2025. 2

[5] Sizhe Lester Li, Evan Kim, Xingjian Bai, Tong Zhao, Tao Pang, Max Simchowitz, and Vincent Sitzmann. Turning video models into generalist robot policies. *arXiv preprint arXiv:2605.27817*, 2026. 2

[6] Qwen Team. Qwen3-VL-4B-Instruct model card. `https://huggingface.co/Qwen/Qwen3-VL-4B-Instruct`, 2025. 4

[7] Stephane Ross, Geoffrey Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, pages 627–635, 2011. 3

[8] Kohei Sendai, Maxime Alvarez, Tatsuya Matsushima, Yutaka Matsuo, and Yusuke Iwasawa. Leave no observation behind: Real-time correction for VLA action chunks. *arXiv preprint arXiv:2509.23224*, 2025. 2

[9] Michael Tschannen, Alexey Gritsenko, Xiao Wang, Muhammad Ferjad Naeem, Ibrahim Alabdulmohsin, Nikhil Parthasarathy, Talfan Evans, Lucas Beyer, Ye Xia, Basil Mustafa, Olivier Hénaff, Jeremiah Harmsen, Andreas Steiner, and Xiaohua Zhai. SigLIP 2: Multilingual vision-language encoders with improved semantic understanding, localization, and dense features. *arXiv preprint arXiv:2502.14786*, 2025. 4

[10] Jianwei Yang, Reuben Tan, Qianhui Wu, Ruijie Zheng, Baolin Peng, Yongyuan Liang, Yu Gu, Mu Cai, Seonghyeon Ye, Joel Jang, Yuquan Deng, and Jianfeng Gao. Magma: A foundation model for multimodal AI agents. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14203–14214, 2025. 2

[11] Brian Zhu, Momen Khalil, E Harrison, Emanuele Poggi, Philipp Schmitt, Bernd Kast, Philine Meister, Pranav Atreya, Qiyang Li, Finn Ferchau, Cesar Colmenero, Yash Shahapurkar, Gokul Narayanan, Melih Erdogan, Kai Wurm, Georg von Wichert, Oier Mees, Eugen Solowjow, Andrew Wagenmaker, and Sergey Levine. Learning to act while waiting: RL finetuning of generalist robot policies under inference latency. *arXiv preprint arXiv:2608.23831*, 2026. 2

[12] Yixiang Zhu, Yonghao Chen, Jiaxiang Zou, Yusong Hu, and Xinyu Chen. DEFLECT: Temporal counterfactual preference learning for delay-robust asynchronous VLAs. *arXiv preprint arXiv:2605.19294*, 2026. 2

LLM Usage Statement

Generative AI tools assisted with research discussion, implementation, experiment orchestration, literature retrieval, data analysis, and manuscript drafting. Astra and Qwen3-VL are also components evaluated in the research. Numerical results in this draft are derived from recorded experiments. Unfinished experiments are explicitly marked TODO. The authors must verify the final text, references, code, and results and remain responsible for the submitted work.